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Code a 2D Mobile Game in less than 6 hours



Learn How to Design and Code a 2D Mobile Game with Flame Engine and Flutter.

Follow along!

Created by

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27 backers pledged CA\$ 1,655 to help bring this project to life.

 **Last updated** July 12, 2022

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CA\$ 1,655

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27

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Story

Have you ever dreamed of creating your own mobile game? Have you ever wanted to code your own mobile game? If the answer is **YES**, then **THIS course is for you.**

You can **learn to code a 2D mobile game in LESS than 6 hours**

Welcome to the *Flame Engine 2D Game Crash Course* where you will learn how to code a full mobile game in **less than 6 hours.**



There are many **game enthusiasts** out there who would love to learn how to write mobile games but do not necessarily have the time to dedicate to it.

Many Game Engines out there have a steep learning curve so creating a full mobile game would take considerable time and effort.

We have created this course for all the coders who want to get into creating their own mobile game using a 2D engine, and more importantly, we have created this course to take you from 0 to hero in less than 6 hours.

Why Flame Engine

The Flame Engine is a **minimalist** 2D game engine that runs on top of the **Flutter** framework. So it is light-weight with a small footprint.

This is perfect for mobile games.

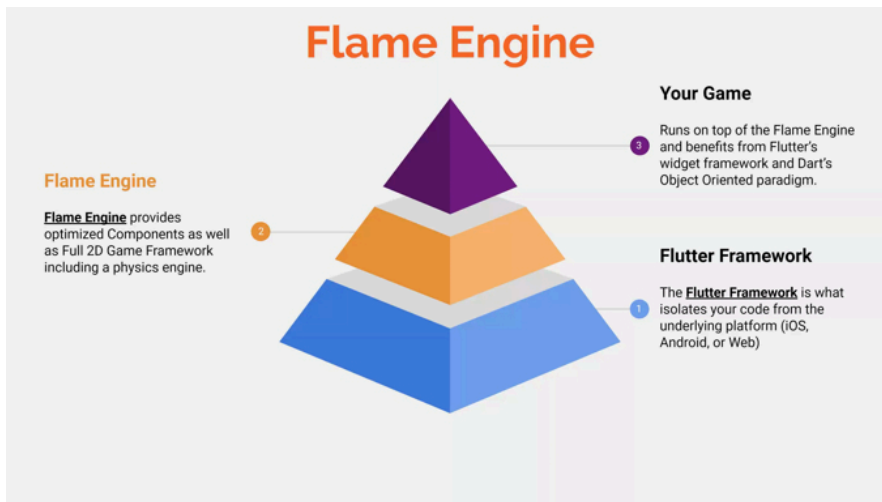


Figure 1 - The Flame Engine

Advantages of The Flutter Framework

One code base and *multiple platform deployment*. Basically write once, and deploy to both **iOS** and **Android**. This is a huge advantage as it saves you time and effort.

And this is great for mobile games since we do not rely on the specific OS GUI as we basically take over the screen and create our own GUI from scratch.

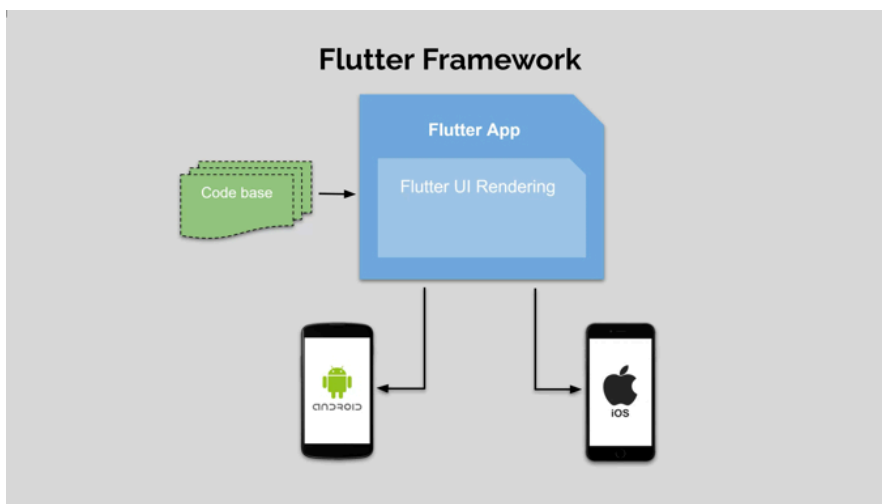


Figure 2 - The Flutter Framework

Crash Course Format

This is a crash course. So we remove all the fluff from the usual approach and we focus and zone in, with laser precision, on the fundamentals that apply to all 2D engines. We explain the necessary fundamentals such as:

- Game Loop
- Component Rendering
- Spritesheet Animation
- Game Mechanics
- Inter-Component Communication and Messaging.
- Component Interoperability And Composability
- Collision Detection
- Particle Physics
- Math for Game Developers such as:
 - Vector math
 - Velocity control
 - Rotational control
 - Collision Detection
 - Boundary Detection
- Game Parallax - For creating complex background motion.
- Sound Design - For adding sounds, sound effects, and music to your game
- We also explore all the needed APIs in detail.

And in the end, we use all this to develop a famous retro-game.

Focus on Fundamentals

We will spend the first 2 hours focusing on fundamentals with code and architectural examples. Below are some stills from the course dealing with fundamental aspects of the Game Engine.

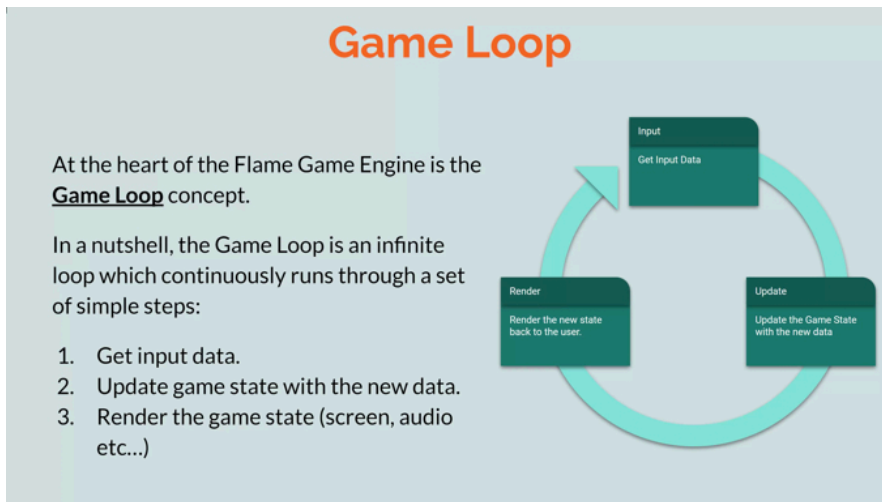
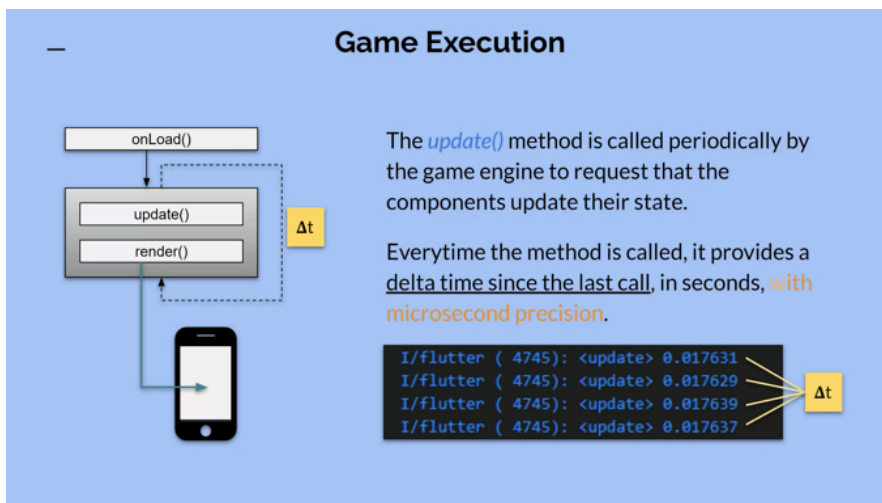


Figure 3 - Game Loop

Figure 4 - Game Execution with `update()` method

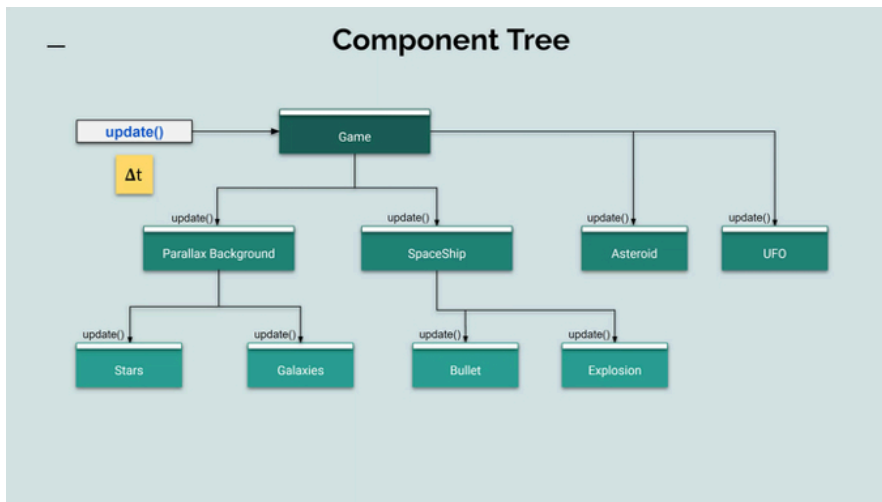


Figure 5 - Component Tree Hierarchy

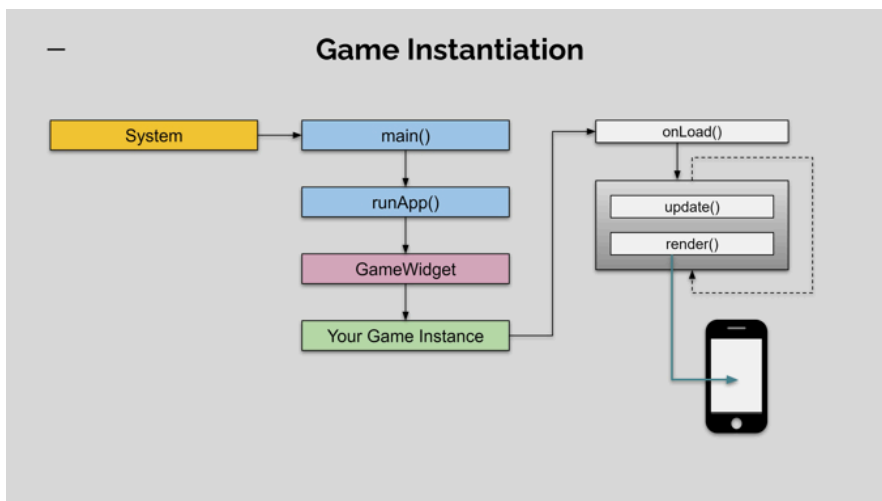


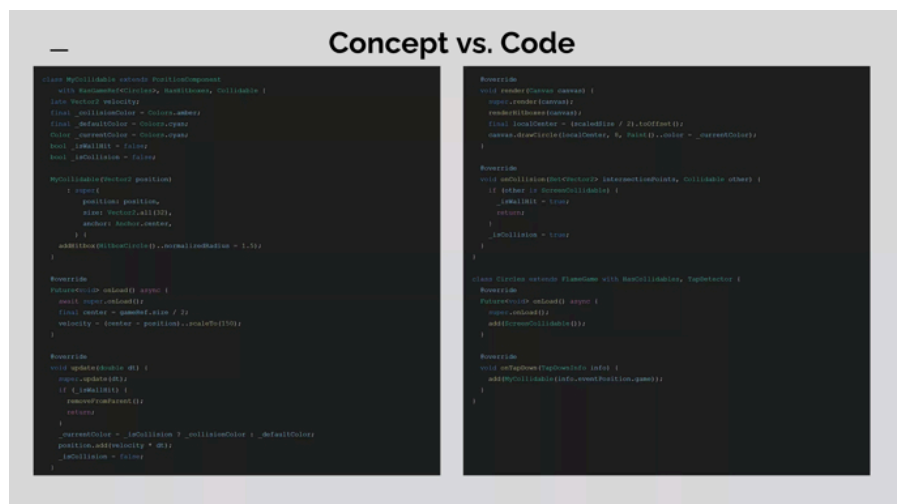
Figure 6 - Game Widget Instantiation lifecycle.

Architectural Approach

Our approach to teaching the material is a bit different from most courses. We take the *architectural approach*. This means that we will also teach you some **Software Engineering** while we are teaching you about the Flame Engine.

We will use **UML** diagrams to show you the context of what we are doing. All of this is to give you a better perspective of how everything fits

So we will not just be doing this:



```

classDiagram
    class Circles
    class MyCollidable
    class Canvas
    class FlameGame
    class PositionComponent
    class ScreenCollidable
    class VapDownInfo
    class HasCollidables
    class TapDetector
    class HasGameRefCircles
    class HBoxCircle
    class Collidable
    class HasHBoxCircles
    class Vector2
    class Paint
    class Anchor
    class Colors

    Circles "1" --> "0..*" MyCollidable
    MyCollidable "1" --> "0..*" Canvas
    Circles "1" --> "0..*" PositionComponent
    Circles "1" --> "0..*" ScreenCollidable
    Circles "1" --> "0..*" VapDownInfo
    Circles "1" --> "0..*" HasCollidables
    Circles "1" --> "0..*" TapDetector
    Circles "1" --> "0..*" HasGameRefCircles
    Circles "1" --> "0..*" HBoxCircle
    Circles "1" --> "0..*" Collidable
    Circles "1" --> "0..*" HasHBoxCircles
    Circles "1" --> "0..*" Vector2
    Circles "1" --> "0..*" Paint
    Circles "1" --> "0..*" Anchor
    Circles "1" --> "0..*" Colors
    
```

Figure 8 - Bouncing Balls Simplified UML diagram

In other words **we will not only work with code**. The main objective here is to teach you to think like a Software Architect. We do not want you to just code this game. We want you to go through the process of designing and architecting it in a professional manner.

Using Case Studies

Part of how we teach you in this course revolves around **practical case studies**. Each case study shows you a full solution to a given element of 2D game coding such as for example Joystick control or collision detection.



Figure 10 - Joystick control case study



Figure 11 - Collision detection case study

The Game we will develop

We will develop, from scratch, using our architectural approach the famous retro-game from ATARI: Asteroids.

This is a perfect game for learning all the fundamentals.



Figure 9 - ATARI's Asteroids game still.

Why this particular game

This is a perfect game to learn a 2D game engine for the following reasons:

- It has great game mechanics such as:
 - Spaceship control
 - Spaceship explosion
 - Tracking of points and player life
 - Firing of bullets
 - Asteroid motion/rotation/(and splitting up)
 - Occasional Bonus UFO showing up.
- This will lead us to fully explore and understand:
 - Joystick control
 - Game Loop update and render
 - Timer component callbacks
 - Vector Math
 - Collision Detection
 - Particle Physics
 - Refresh Rate independent rendering
 - Sound design and game music
 - Added parallax effects for background

This game is also great to be played on both a phone as well as a tablet device.

Assumptions

We assume that you have *some coding experience* and we also assume that you know Flutter/Dart and have worked with it.

But in case you have not worked with Flutter do not fret!

The course is designed to help you with the setup and a solid understanding of the language elements because of the architectural approach.

So in reality all you need is experience with an Object Oriented language such as **Java**, **Swift**, **C#**, **C++**, or **TypeScript** and you will be fine!

Why we launched this Campaign

We ask you for your pledges as **we would love to create content that would benefit anybody who wants to learn how to code a full 2D mobile game** but has limited time on their hands (don't we all?)

We need to polish the course, render it and optimize the game code and we will be done, with your help that is!

Games offer an escape, and we love that facet of game development, so we would love to show others how even with limited time you can focus your energy to create a mobile game of your own.

Isn't there a game developer in all of us?

Risks and challenges

We have a lot of experience teaching, mentoring, as well as developing and architecting software solutions, including game development. So we are confident that we will deliver this course to you and on time. We do not envision any issues or delays.

Since this is an electronic medium (i.e. videos content, PDF content, as well as code) there are no issues associated with shipping (like delays)
This course will be available to you for download.

The only challenge is of course doing a good job of giving you well curated content, so that you truly can learn as much as you possibly can in less than 6 hours. We have really tried to distill the essence of 2D game development to give to you.

And this, indeed, is our mission statement, to provide and impart the most valuable content in the least amount of your time. We hope that we have succeeded.

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